A Near-Linear-Time Solver for Bilevel Congestion Network Design via Directed Nonlinear Laplacian Flow

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Bilevel traffic network design—optimizing tolls or capacities over a lower-level user equilibrium—is nonconvex and NP-hard; every practical method requires, at each outer step, an equilibrium solve and an equilibrium sensitivity. The sensitivity has been near-linear in the number of design variables since Tobin and Friesz (1988), but computing it requires factorizing the equilibrium Jacobian, and that factorization grows superlinearly with network size, capping exact design methods near a hundred links while forward assignment reaches metropolitan scale. We remove that bottleneck. The directed user equilibrium with separable costs is a source-form nonlinear Laplacian flow (NLF) $B\rho(B^\top\phi - \tau) = d$ whose Newton Jacobian, despite the one-sided rectified edge law, remains a symmetric weighted graph Laplacian on the active subgraph. Both the forward equilibrium and the adjoint design gradient therefore reduce to near-linear-time Laplacian solves, via a smoothing homotopy that handles the rectified law's ill-conditioning. On Sioux Falls, NLF matches Frank–Wolfe and finite-difference sensitivities, and toll design matches the best scaled marginal-cost toll. Confirmed on a corpus of 96 real irregular graphs up to 6×10^6 arcs, NLF's wall-clock scales near-linearly while direct factorization scales superlinearly and exhausts memory at large scale; NLF wins beyond about 10,000 arcs. The advantage is specific to poorly-separable networks (e.g., selfish routing); on near-planar road networks, nested dissection is already near-optimal. We extend the framework to multicommodity logit stochastic user equilibrium, where a block-diagonal preconditioner solves the coupled Newton system in single- to low-double-digit iterations and the adjoint returns all toll gradients in one solve; on Sioux Falls and Anaheim, destination-bundled design cuts total system travel time by 7.7–13.1%. Code is available at `https://github.com/orenlivne/dnlf`.

Key words: network design; traffic assignment; user equilibrium; bilevel optimization; nonlinear Laplacian flow; algebraic multigrid; adjoint sensitivity; congestion pricing

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1. Introduction

Given a road network with m directed links and travel-time (latency) functions $t_a(f_a)$ that increase with flow, the *lower-level* user equilibrium (UE) assigns origin–destination demand so that no traveler can lower their experienced cost by switching route (Wardrop's principle). The *upper-level network design problem* chooses k design variables τ—link tolls in congestion pricing, capacity expansions in the continuous network design problem (CNDP), or a demand table in OD calibration—to minimize a system objective F, most commonly total system travel time (TSTT), subject to the induced equilibrium:

$$\min_{\tau\in\mathcal{T}} F\big(f^\star(\tau)\big) \qquad \text{s.t.} \qquad f^\star(\tau) \text{ solves the UE at design } \tau. \tag{1}$$

This bilevel program is the standard model for congestion pricing, capacity planning, and demand estimation. It is nonconvex, nondifferentiable, and *NP*-hard even with affine latencies (Gairing et al. 2017); second-best tolling is *NP*-hard as well (Hoefer et al. 2008). Practical methods therefore seek a local (KKT) stationary point, and essentially all of them—sensitivity-analysis-based (SAB) descent (Tobin and Friesz 1988, Josefsson and Patriksson 2007, Chiou 2005), mathematical programs with equilibrium constraints, and metaheuristics—run an *outer* loop where each iteration solves the lower-level equilibrium and then a sensitivity (equilibrium derivative).

1.1. Computational cost, and where it lives

The design gradient itself is cheap in the number of design variables: since Tobin and Friesz (1988), $\nabla_\tau F$ is obtained by factorizing the equilibrium Jacobian *once* and back-substituting one right-hand side per design variable, so its cost is independent of k (an adjoint / implicit-function computation (Josefsson and Patriksson 2007, Amos and Kolter 2017, Blondel et al. 2022)). The bottleneck is that this factorization and the lower-level solve that precedes it are direct sparse factorizations whose cost grows superlinearly (fill-dominated) in the active-path/subnetwork size, or a slowly converging Frank–Wolfe assignment. This is why exact design methods in the literature stall near Sioux Falls scale (76 links) while origin-based forward assignment routinely reaches $40,000$–$75,000$ links (Bar-Gera 2002, Dial 2006). The gap between forward and design scalability is one to two orders of magnitude, and is purely a linear-algebra gap.

1.2. Contribution

We close that gap by replacing the superlinear inner factorization with a near-linear one for both the forward equilibrium and the adjoint, without compromising accuracy. Our contributions are:

1. **A Laplacian formulation of the directed equilibrium (Section 2).** We write the separable directed UE as a source-form nonlinear Laplacian flow (NLF) $B\rho(B^\top\phi-\tau)=d$ with a one-sided (rectified) edge law, and show that the Newton Jacobian is a *symmetric* graph Laplacian on the active subgraph (Proposition 1). Orientation and one-sidedness do not break the Laplacian; they only determine which arcs are active. Hence the near-linear Laplacian toolbox applies.
2. **A near-linear equilibrium solver robust to the dead zone (Section 3).** The rectified law is nonsmooth: a link carries flow only once its potential drop exceeds the free-flow cost t^0_a, so the active set—and the conditioning of the Jacobian—changes during a solve, defeating a naively frozen multigrid hierarchy. We introduce a *smoothing homotopy*: a softplus dead-zone

width δ that keeps the conductance strictly positive and the linearized system uniformly well conditioned, continued from a smooth problem to the exact law, with one Algebraic Multigrid (AMG) setup per δ-level and a final exact-law polish. The number of setups is bounded independently of graph size.

3. **A near-linear adjoint and design loop (Section 4).** The design gradient is one additional Laplacian solve with the same operator; projected-gradient descent on it drives TSTT monotonically to the stationary point.
4. **Validation (Section 5).** On Sioux Falls NLF matches Frank–Wolfe to 10^{-11}; the adjoint matches finite differences to finite-difference accuracy; toll design matches the best scaled marginal-cost (Pigouvian) toll to within near-stationarity. Setups per solve are bounded in graph size and the wall-clock scales near-linearly in m, so design inherits the near-linear inner cost.

1.3. Scope: Irregular Networks

The lower-level equilibrium Eq. (3) is the Wardrop equilibrium of a nonlinear congestion game (Rosenthal 1973, Roughgarden and Tardos 2002), and the design problem Eq. (1) is the optimal-tolling / Stackelberg problem of reducing the price of anarchy (Swamy 2012, Korilis et al. 1997). This model describes selfish routing not only in road traffic but in communication and overlay networks such as multipath and elastic-traffic routing games in the Internet, data-center fabrics, and peer-to-peer overlays (Orda et al. 1993, Larroca and Rougier 2013). The distinction matters for scalability: road networks are near-planar, where a direct nested-dissection factorization of the equilibrium Jacobian is already near-optimal. Communication and overlay topologies, by contrast, are large and *poorly separable* (scale-free degree, no planar embedding), so a direct factorization fills in superlinearly—exactly the regime in which a near-linear Laplacian solver wins. Our scaling study is therefore on irregular networks; for the time we focus on establishing the near-linear advantage only here. The most literal instance is selfish *overlay* routing, where end hosts genuinely choose paths (Qiu et al. 2003), on Internet AS- and router-level graphs that are provably scale-free (Faloutsos et al. 1999). Production traffic engineering optimizes operator-set link weights or bandwidth allocations by linear programming and local search (Fortz and Thorup 2000), not by pricing a selfish equilibrium, and packet-level congestion control is a distinct (utility-maximization) equilibrium. Our contribution is thus methodological, a scalable solver for the well-established equilibrium/design model on standard large irregular topologies, complementing rather than replacing operator tools.

The construction reuses the NLF solver (Livne 2026b) and its near-linear inner engine, Lean Algebraic Multigrid (LAMG+) (Livne 2026a) (approximate Cholesky (AC) (Kyng and Sachdeva 2016) is interchangeable and has a smaller hidden constant but is less robust here). We treat both the single-commodity directed equilibrium (Sections 2 to 5) and its multicommodity logit stochastic-user- equilibrium extension (Section 6), where a block-diagonal preconditioner keeps the inner solve to a handful of iterations. Active-set deflation that would support a single hierarchy valid across all design steps (reducing setups from $\mathcal{O}(1)$-per-solve to $\mathcal{O}(1)$ total) is an open problem (Section 7).

1.4. Related work

Bilevel network design and sensitivity. CNDP and second-best tolling are bilevel/MPEC problems, nonconvex and *NP*-hard (Gairing et al. 2017, Hoefer et al. 2008); surveys (Farahani et al. 2013) catalog SAB descent, MPEC, and metaheuristic methods, with second-best tolling treated by Lawphongpanich and Hearn (2004) and the economic theory of road pricing by Yang and Huang (2005). Equilibrium sensitivity originates with Tobin and Friesz (1988) and was sharpened by Qiu and Magnanti (1989) and by Patriksson and co-authors (Josefsson and Patriksson 2007), who observed that the sensitivity problem is itself a linearized equilibrium solve. Our adjoint Eq. (4) is precisely that observation; the novelty here is not the k-independent gradient but the *near-linear* operator that computes it and the forward solve. The same bilevel problem is studied in algorithmic game theory as optimal tolling and Stackelberg routing to reduce the price of anarchy of a congestion game (Roughgarden and Tardos 2002, Swamy 2012, Korilis et al. 1997, Cole et al. 2003), and in communication networks as competitive / selfish routing (Orda et al. 1993, Larroca and Rougier 2013); that literature focuses on existence, bounds, and mechanism design rather than on a scalable solver for the induced equilibrium, which is our concern. Recent differentiable-bilevel methods (Li et al. 2022, Goyal and Lamperski 2023) break the cold-resolve-per-step pattern with warm-started, truncated first-order inner loops and unrolled/implicit differentiation, but use no multigrid and are demonstrated only at Sioux Falls–Chicago-Sketch scale. Their defining component is a first-order inner equilibrium solve; we compare head-to-head against a strong instance of it (FISTA on the same convex Beckmann energy) in Section 5.

Near-linear Laplacian solvers and nonlinear flow. Nearly-linear-time solvers for symmetric diagonally dominant systems (Kyng and Sachdeva 2016, Livne 2026a) underpin web-scale spectral graph computation but have not, to our knowledge, been used as the inner solve of a bilevel traffic-design loop. The nonlinear Laplacian flow framework (Livne 2026b) unifies max-flow, congestion, and min-delay routing as one monotone edge-law equation and supplies the chord-Newton machinery we build on; the present work extends it from the undirected relaxation to the directed (rectified) equilibrium and to the design (adjoint) problem.

2. The directed equilibrium is a nonlinear Laplacian flow

Let $B \in \mathbb{R}^{n\times m}$ be the node–arc incidence matrix of the directed network, $B_{\cdot a} = e_{t(a)} - e_{h(a)}$ for arc a from tail $t(a)$ to head $h(a)$. Separable BPR latencies are $t_a(f) = t_a^0\big(1+\beta_a(f/\kappa_a)^{p_a}\big)$ for $f \geq 0$. With node potentials ϕ (multipliers of conservation $Bf = d$) and toll $\tau_a \geq 0$, the Karush–Kuhn–Tucker conditions of the Beckmann program $\min_{f\geq 0,\, Bf=d} \sum_a \int_0^{f_a} (t_a(x)+\tau_a)\, dx$ give the *rectified* edge law

$$f_a = \rho_a(g_a - \tau_a), \qquad g_a := (B^\top \phi)_a = \phi_{t(a)} - \phi_{h(a)}, \qquad \rho_a(u) = \begin{cases} t_a^{-1}(u), & u > t_a^0, \\ 0, & u \leq t_a^0, \end{cases} \tag{2}$$

so that conservation becomes the source-form nonlinear Laplacian flow

$$B\,\rho(B^\top \phi - \tau) = d. \tag{3}$$

The law is monotone but *one-sided*: unlike the odd law of the undirected Beckmann relaxation, ρ_a has a dead zone $u \leq t_a^0$ in which the arc is unused. This dead zone is exactly what distinguishes the directed equilibrium from its undirected relaxation.

PROPOSITION 1. *For separable costs the Newton Jacobian of Eq.* (3) *is a symmetric, positive semidefinite weighted graph Laplacian supported on the active arcs. Explicitly, with* $\rho'_a \geq 0$,

$$J = B\,\mathrm{diag}(\rho')\,B^\top = \sum_a \rho'_a\,(e_{t(a)} - e_{h(a)})(e_{t(a)} - e_{h(a)})^\top, \quad \rho'_a = \tfrac{1}{t'_a(f_a)} > 0 \textit{ (active)}, \;\; \rho'_a = 0 \textit{ (else)}.$$

Proof. Differentiating Eq. (3) in ϕ gives $J = B\,\mathrm{diag}(\rho'(B^\top\phi - \tau))\,B^\top$. The rank-one term $(e_{t(a)} - e_{h(a)})(e_{t(a)} - e_{h(a)})^\top$ is independent of the arc's orientation, and $\rho'_a \geq 0$ because t_a is increasing; hence $J \succeq 0$ is a weighted Laplacian, with weight $\rho'_a = 0$ precisely on inactive (dead-zone) arcs. Q.E.D.

Proposition 1 is the structural fact this paper rests on: directedness enters only through *which* arcs are active (a routing boundary), not through the type of the linear operator. Every linearized system is a graph Laplacian, which can be solved fast. This is complete for the single-commodity equilibrium (a general balanced demand, possibly multi-source/multi-sink). The multicommodity case ($f_a = \sum_k f_a^k$ with a cost of the total flow) yields a coupled *block* system—per-commodity Laplacians with a rank-one commodity coupling per arc—which a block-diagonal preconditioner still solves in a handful of iterations; we develop this in Section 6 (Proposition 4).

3. A near-linear equilibrium solver

We solve Eq. (3) by damped chord-Newton on the frozen-linearization Jacobian J, delegating each linearized solve to a near-linear Laplacian engine and globalizing with the convex Beckmann energy $E(\phi) = \sum_a \Psi_a\big((B^\top\phi)_a - \tau_a\big) - d^\top\phi$, $\Psi'_a = \rho_a$, exactly as in the NLF framework (Livne 2026b). Two features of the directed problem require care.

3.1. Ill-conditioning and the active set

On inactive arcs $\rho'_a = 0$, so J is singular unless a floor is imposed; a tiny floor $\varepsilon \max_a \rho'_a$ with $\varepsilon = 10^{-12}$ keeps J connected without perturbing the Newton step, but drives the condition number to $\kappa \sim \varepsilon^{-1}$. More importantly, as flow builds the active set changes: arcs cross the free-flow threshold t^0_a, so the operator that an AMG hierarchy is built for is stale a few Newton steps later. A hierarchy frozen across the whole solve stalls short of the equilibrium. (Conceivably, the hierarchy could be updated only locally, around active set changes; this is left for future work.)

3.2. Smoothing homotopy

We remove the dead-zone nonsmoothness by a homotopy in a width $\delta > 0$. Replace the excess $e = g - t^0_a$ by $\delta\,\mathrm{softplus}(e/\delta)$, which is smooth, strictly positive, and tends to $\max(e, 0)$ as $\delta \to 0$; the resulting smoothed law $\rho_{a,\delta}$ has $\rho'_{a,\delta} = \sigma(e/\delta)/t'_a(f_a) > 0$ *everywhere* (where σ is the logistic function), so J is a full, uniformly well-conditioned Laplacian with no dead zone. We continue δ geometrically with a fine ratio (0.85) from $\delta_0 = \frac{1}{2}\bar{t}^0$ down to $\delta_L \approx 8 \times 10^{-9}\bar{t}^0$, building one AMG hierarchy at the start of each δ-level and freezing it within the level. The fine ratio is what makes the solve robust: each level is then a small, well-conditioned perturbation of the last, so the frozen hierarchy stays valid and a few warm-started chord-Newton steps drive that level to the target tolerance. A short final polish on the exact law Eq. (2), warm-started from the δ_L solution (already within $O(\delta_L)$ of the exact equilibrium), recovers the exact rectified equilibrium (Algorithm 1). A coarse schedule, or solving only the last level tight, instead leaves each tight solve far from converged on the ill-conditioned activation Jacobian, where the frozen hierarchy stalls short of machine precision. Because the number of δ-levels is fixed and each is a small perturbation, the total number of inner-engine setups (one build per δ-level that requires a Newton step) is bounded independently of graph size m; solving intermediate levels only loosely ($\varepsilon_{\mathrm{int}}$) is a faster variant that reaches $\sim 10^{-3}$, sufficient for design (Section 5).

Robustness on hub-dominated graphs. The loose-intermediate variant keeps every homotopy level cheap and defers the single tight solve to the exact-law polish. On most graphs a frozen-hierarchy polish suffices; on extreme hub graphs (dense web crawls, where a small densely-linked cluster concentrates the smoothed conductance) the one-sided law drives the active-subgraph Laplacian near-singular—an isolated, low-rank near-null mode (condition number $\sim 10^{12}$; its eigenvector is essentially uniform over the offending cluster, which we verified by inverse power iteration)—on which a frozen tight solve stalls short of tolerance. Rebuilding the hierarchy adaptively during the polish, whenever the residual ceases to fall, clears the stall. The corpus study (Section 5.5) uses this loose-homotopy, adaptive-polish combination: it converges every one of 96 real graphs (median residual 8×10^{-5}; only the most hub-dominated crawls stop at $\sim 10^{-3}$ within budget), at the cost of a modestly larger, conditioning-dependent setup count. A mode-deflating polish that would restore machine precision on those few graphs is future work.

We give a self-contained convergence guarantee for the smoothed level solve, so the paper does not depend on external convergence theory.

PROPOSITION 2 **(global convergence of the smoothed level solve)**. *Fix $\delta > 0$ and tolls τ. The smoothed Beckmann energy $E_\delta(\phi) = \sum_a \Psi_{a,\delta}\big((B^\top\phi)_a - \tau_a\big) - d^\top\phi$, with $\Psi'_{a,\delta} = \rho_{a,\delta}$, is convex, and strictly convex and coercive on the quotient $\mathbb{R}^n/\langle\mathbf{1}\rangle$ whenever $\sum_a d_a = 0$, the graph is (weakly) connected, and d is* strictly routable *(interior to the cone of demands realizable by strictly positive arc flows—the feasibility a Wardrop equilibrium presumes). Its unique minimizer (modulo an additive constant) is the smoothed equilibrium $B\rho_\delta(B^\top\phi - \tau) = d$. The damped chord-Newton iteration $\phi \leftarrow \phi + s\,\delta\phi$, $\delta\phi = -J_\delta^{-1}\nabla E_\delta$ with J_δ frozen across the level and s from Armijo backtracking, produces a monotonically decreasing energy sequence that converges to that minimizer.*

Proof. $\rho_{a,\delta}$ is nondecreasing (its derivative $\rho'_{a,\delta} = \sigma(e/\delta)/t'_a(f_a) > 0$), so each $\Psi_{a,\delta}$ is convex and E_δ is convex with Hessian $J_\delta = B\,\mathrm{diag}(\rho'_\delta)\,B^\top$. Because the smoothed law has *no dead zone*, $\rho'_{a,\delta} > 0$ on *every* arc, so J_δ is a connected weighted graph Laplacian, positive definite on $\mathbf{1}^\perp$, which gives strict convexity of E_δ on the quotient. Coercivity is a separate condition: the one-sided smoothed law saturates ($\rho_{a,\delta} \to 0$ and $\Psi_{a,\delta}$ bounded as the arc excess $\to -\infty$), so $\sum_a \Psi_{a,\delta}$ stays bounded only along directions $v \perp \mathbf{1}$ with $B^\top v \le 0$ (all arc drops non-increasing); on such a direction the linear term $-d^\top\phi$ dominates and $E_\delta \to +\infty$ *precisely because* d is strictly routable, i.e. $d^\top v < 0$ for every nonzero such v (cone duality with the nonnegative-flow feasibility cone). Strict convexity together with coercivity yields a unique minimizer on the quotient, characterized by $\nabla E_\delta = B\rho_\delta(B^\top\phi - \tau) - d = 0$. For the iteration, $J_\delta \succ 0$ on $\mathbf{1}^\perp$ makes $\delta\phi$ a descent direction ($\nabla E_\delta^\top \delta\phi = -\nabla E_\delta^\top J_\delta^{-1}\nabla E_\delta < 0$), and Armijo backtracking on the convex, coercive, smooth E_δ gives sufficient decrease at each step; the standard damped modified-Newton argument for a fixed positive-definite metric then yields convergence to the unique minimizer. Q.E.D.

Proposition 2 covers each smoothed level; the homotopy is a continuation of these strictly-convex problems toward $\delta \to 0$, and the polish is a warm-started solve of the exact (piecewise-smooth, still convex) energy. The main solve tightens *every* level to the target tolerance ε_{tol}, which the fine schedule makes cheap (each level a small perturbation). For *design*, where 10^{-3} suffices, a loose-intermediate variant instead solves each intermediate level ($\ell < L$) only to $\varepsilon_{\text{int}} \approx 3 \times 10^{-2}$ (a handful of chord-Newton steps): by the nested-iteration principle its solution need only warm-start the next level. Over-solving a level one is about to leave is wasted, while under-solving is absorbed by the next level and the polish, so the loose-variant work is insensitive to ε_{int} over more than a decade, and looser tolerances are cheaper.

Algorithm 1 Near-linear directed-UE solve (cold start). A *fine* geometric δ-schedule (ratio 0.85, from $\delta_0 = \frac{1}{2}\bar{t}^0$ down to $\delta_L \approx 8 \times 10^{-9}\bar{t}^0$) makes each level a small, well-conditioned perturbation of the last, so solving *every* level to the target tolerance ε_{tol} (10^{-9} in our runs) reaches machine precision without stalling. A loose-intermediate variant ($\varepsilon_{\text{int}} = 3 \times 10^{-2}$ at intermediate levels) reaches $\sim 10^{-3}$ faster, which is all design requires (Section 5).

$\phi \leftarrow 0$; choose fine geometric schedule $\delta_0 > \cdots > \delta_L$
for $\ell = 0, \ldots, L$ **do** ▷ smoothing homotopy: one hierarchy build per level
  build AMG hierarchy for J_δ at $\delta = \delta_\ell$; run chord-Newton on $B\rho_{\delta_\ell}(B^\top\phi - \tau) = d$ to ε_{tol}, warm-started from level $\ell - 1$ (a few steps, as the level is a small perturbation); hierarchy **frozen** within the level
end for
polish: chord-Newton on the exact law $B\rho(B^\top\phi - \tau) = d$ to ε_{tol}, warm-started from ϕ (within $O(\delta_L)$ of the exact equilibrium, so a few steps)
return ϕ, arc flows $f_a = \rho_a((B^\top\phi)_a - \tau_a)$

4. Bilevel design by a near-linear adjoint

We minimize $F = \text{TSTT}(f) = \sum_a t_a(f_a) f_a$ over tolls $\tau \geq 0$, where $f_a = \rho_a((B^\top\phi)_a - \tau_a)$ at the equilibrium $\phi = \phi^\star(\tau)$.

Proposition 3 **(near-linear adjoint gradient)**. *At a differentiable equilibrium with active-arc conductances $\rho' > 0$,*

$$\nabla_\tau F = \rho' \odot \left(B^\top\lambda - m_c\right), \qquad J\lambda = B\,(m_c \odot \rho'), \tag{4}$$

where $m_{c,a} = t_a(f_a) + f_a t'_a(f_a)$ is the marginal cost and $J = B\,\mathrm{diag}(\rho')\,B^\top$ is the equilibrium Jacobian of Proposition 1. The gradient with respect to all k tolls is thus obtained from a single additional solve with the operator J.

Proof. Write $g = B^\top\phi - \tau$, so $f = \rho(g)$ and $F(\tau) = \sum_a c_a(f_a)$ with $c_a(f) = t_a(f)f$, $c'_a = m_{c,a}$. Differentiating F in τ_b, $\partial_{\tau_b}F = \sum_a m_{c,a}\rho'_a(\partial_{\tau_b}g_a)$ with $\partial_{\tau_b}g_a = (B^\top\phi_{,b})_a - \delta_{ab}$, where $\phi_{,b} = \partial\phi^\star/\partial\tau_b$. Implicitly differentiating the equilibrium $B\rho(B^\top\phi - \tau) = d$ gives $J\,\phi_{,b} = B\,\mathrm{diag}(\rho')\,e_b = \rho'_b\,Be_b$. Hence, with $w := m_c \odot \rho'$, $\partial_{\tau_b}F = w^\top(B^\top\phi_{,b}) - m_{c,b}\rho'_b = (Bw)^\top\phi_{,b} - m_{c,b}\rho'_b = \rho'_b\,(Bw)^\top J^{-1}(Be_b) - m_{c,b}\rho'_b$. Defining the adjoint $\lambda := J^{-1}(Bw)$ (i.e. $J\lambda = Bw$), symmetry of J gives $(Bw)^\top J^{-1}(Be_b) = \lambda^\top(Be_b) = (B^\top\lambda)_b$, so $\partial_{\tau_b}F = \rho'_b\big((B^\top\lambda)_b - m_{c,b}\big)$, which is Eq. (4). The single solve $J\lambda = Bw$ yields the whole gradient. Q.E.D.

This recovers the classical k-independence of equilibrium sensitivity (Tobin and Friesz 1988, Josefsson and Patriksson 2007) but at near-linear rather than superlinear (fill-dominated) cost. On the exact rectified law J is singular off the active subgraph (inactive arcs have $\rho' = 0$, isolating their nodes); Eq. (4) is therefore solved on the active-arc node support (equivalently, with an infinitesimal conductance floor). The gradient is otherwise insensitive to the floor—a floor of 10^{-12} reproduces central finite differences to finite-difference accuracy, whereas 10^{-6} biases it by 5%. Projected-gradient descent $\tau \leftarrow \max(\tau - \eta \nabla_\tau F, 0)$ with a backtracking line search on F, where each trial requires an exact equilibrium solve by Algorithm 1, yields a monotone design method (Algorithm 2).

Algorithm 2 Bilevel toll design

$\tau \leftarrow 0$; $(\phi, f) \leftarrow \textsc{Solve}(\tau)$ ▷ Algorithm 1
for $k = 1, 2, \ldots$ until stationary **do**
    $g \leftarrow \nabla_\tau F$ from Eq. (4); $\eta \leftarrow \eta_0$
    repeat ▷ backtracking on the exact objective
        $\tau' \leftarrow \max(\tau - \eta g, 0)$; $(\phi', f') \leftarrow \textsc{Solve}(\tau')$; $\eta \leftarrow \eta/2$
    until $F(f') < F(f)$
    $(\tau, \phi, f) \leftarrow (\tau', \phi', f')$
end for

5. Numerical results

We refer to our directed-equilibrium solver (Algorithm 1, a smoothing-homotopy continuation with a near-linear inner engine) as NLF throughout. We use the Sioux Falls benchmark ($n = 24$, $m = 76$) and size-scaled grid networks. The inner engine is LAMG+ (Livne 2026a) by default; AC (Kyng and Sachdeva 2016) is interchangeable, tracking it within ≈1.5× and fitting the same near-linear scaling (the direct-baseline head-to-head in Table 1 times AC).

5.1. Correctness of the Equilibrium

On a congested Sioux Falls instance (29 active arcs, flow split across paths) Algorithm 1 matches an independent Frank–Wolfe assignment to a relative flow difference of $4\text{–}9 \times 10^{-11}$ across origin–destination pairs, i.e. to Frank–Wolfe's own accuracy. The number of inner-engine setups is ≈30 here—one AMG build per δ-level that requires a Newton step, fewer than the full schedule because easy late levels are already converged—rising to ≈100 on the stiffer aggregate-demand design (Section 4) and size-scaled solves (Table 1). In every case it is bounded independently of graph size.

5.2. Correctness of the Adjoint

On the same congested instance the adjoint gradient Eq. (4) matches central finite differences of TSTT to a maximum relative error of 2.6×10^{-3} over all active arcs (limited by the finite-difference step and equilibrium tolerance, not by the adjoint). The tiny conductance floor $\varepsilon = 10^{-12}$ is essential: a larger floor (10^{-6}) biases the adjoint to a 5% error.

5.3. Design Optimization

Projected-gradient toll design (Algorithm 2) reduces TSTT monotonically from 9.633×10^5 to 9.137×10^5, a 5.15% reduction over 35 accepted steps, with every equilibrium solved exactly (relative flow error $\sim 10^{-9}$). This matches a scaled marginal-cost (Pigouvian) toll sweep, which bottoms out at 5.12%; the gradient attains a marginally lower TSTT than the best scaled Pigouvian toll, consistent with near-stationarity. Solving every equilibrium only to a loose 10^{-3} tolerance reaches the *same* optimum (TSTT within 0.004%, tolls within $\|\Delta\tau\| = 0.13$), confirming that the design gradient is insensitive to intermediate solve accuracy—so the faster loose-intermediate continuation costs nothing in design quality. The residual gap reflects where the loose tolerance was applied, not where it was measured: it fixes the accepted τ trajectory itself, so tightening only the *final* equilibrium evaluation to ε_{tol} would sharpen the reported TSTT without moving τ, and closing the gap in τ would instead require a design-loop polish—a few tight-tolerance steps from the loose optimum, mirroring Algorithm 1's polish stage—which we expect to shrink it only marginally, since the loose run already sits within the $\sim 10^{-6}$ second-order neighborhood of stationarity. The reduction is modest because a single origin–destination pair admits little rerouting; the design problem is more involved for the multicommodity instances of Section 6.

Design at scale. Exact sensitivity-analysis (SAB) design factorizes the equilibrium Jacobian at every step (superlinearly, fill-dominated). The reported literature (Tobin and Friesz 1988, Josefsson and Patriksson 2007, Chiou 2005) caps this near $\sim 10^2$ links, but the operative modern bound is the memory wall of the fill itself: even the fair CHOLMOD+METIS factorization we measure in Section 5 exhausts RAM at $m \approx 6.4 \times 10^5$. The question is whether the *full* design loop of Algorithm 2—not merely one gradient—runs at metropolitan-communication scale. We run it to convergence on the `p2p-Gnutella31` overlay ($n = 62{,}561$, $m = 2.96 \times 10^5$; 324× Anaheim's link count), a graph in the paper's selfish-overlay-routing regime, at a demand calibrated to heavy congestion (peak volume/capacity ≈10). Projected-gradient tolling takes 20 accepted steps to a monotone 0.56% TSTT reduction; each step is *one* adjoint—returning the gradient with respect to *all* 2.96×10^5 tolls in ≈0.1–0.4 s via a single near-linear Laplacian solve (Proposition 3)—plus *one* warm equilibrium re-solve at ≈10–30 s, for ≈20 min total including the cold solve. Each equilibrium is solved to the design-grade 10^{-3} tolerance (the achievable residual on this hub-dominated overlay, Section 3); the loose-vs-tight insensitivity established on Sioux Falls (final TSTT within 0.004%, Section 4) is two orders below the 0.56% reduction, so the reduction is not solve noise. The two operations an SAB step performs by direct factorization are here both near-linear, so a design iteration at 3×10^5 links costs seconds. The single sub-second adjoint scales further still: on the 1.22-million-arc CAIDA router graph one adjoint returns all 1.22×10^6 toll gradients in 0.9 s. As with the single-OD Sioux Falls case (5.15%), the reduction here is modest because a single fungible commodity admits limited rerouting; the larger, model-calibrated reductions (7.7–13.1%) come from the multicommodity design of Section 6. The point is that the design *loop*, unchanged, runs two-to-three orders of magnitude beyond that literature-reported SAB ceiling; we do not run an SAB outer method here, so this is a linear-algebra-kernel comparison (near-linear vs. superlinear per step), not a full head-to-head.

5.4. Widening Advantage on Size-Scaled Networks

On size-scaled irregular (poorly-separable) congested networks the fine-continuation solve reaches a 10^{-9} equilibrium residual with a wall-clock that scales near-linearly, $t \propto m^{1.17}$ (Table 1, Fig. 2). We compare against two direct factorizations of the pinned Laplacian J, each run under the identical continuation with the same frozen-per-level discipline as the multigrid hierarchy. The first is the unsymmetric UMFPACK/COLAMD LU used by the sensitivity-analysis incumbents; because Proposition 1 makes J symmetric positive definite, we also run the fair, stronger direct incumbent—a CHOLMOD supernodal Cholesky under a METIS nested-dissection ordering. Both are superlinear: UMFPACK fits $m^{1.41}$, and CHOLMOD+METIS $m^{1.24}$ overall ($m^{1.43}$ once the factorization dominates, $m \geq 8 \times 10^4$). The nested-dissection Cholesky is the stronger direct solver—faster than UMFPACK at all but one (anomalous, $m{=}4 \times 10^4$) size (147 vs. 273 s at $m{=}1.6 \times 10^5$) and reaching twice as far before its memory wall—yet NLF is faster than both beyond the smallest, setup-dominated point (where the three agree to within $\approx$2%); the margin over the fair baseline is 1.12× at $m{=}1.6 \times 10^5$ and 1.79× at $m{=}3.2 \times 10^5$ (a single run; a 3-seed synthetic re-timing gives exponent range $[1.06, 1.23]$, so the largest-m point margin is noisy), so the decisive evidence is the exponent gap—near-linear vs. $m^{1.43}$—and the memory wall, not the point margin. Both direct solvers exhaust RAM where NLF does not: UMFPACK at $m{\approx}3.2 \times 10^5$ and CHOLMOD+METIS at $m{\approx}6.4 \times 10^5$, while NLF completes $m{=}6.4 \times 10^5$ in $\approx$730 s and scales to 6×10^6 arcs on the corpus (Section 5.5). The near-linear-vs.-superlinear separation and the memory wall are therefore properties of the fill on these weak-separator graphs, not artifacts of the COLAMD ordering: the fair nested-dissection Cholesky shifts the wall out by one size doubling but does not remove it. The near-linear exponent is inherited from the inner solve, not the outer iteration: the total chord-Newton step count ($\approx$1600), inner-engine setups ($\approx$100, one AMG build per fine δ-level), and per-step Jacobian assembly are all graph-size-independent. Across three random-graph seeds the runtime is $\approx O(m^{1.10})$, consistent with the 96-graph corpus fit $t \propto m^{1.10}$ (Section 5.5). We observed a run-to-run spread reflecting single-machine memory-bandwidth variance at the largest sizes, so the exponent is a wall-clock measurement rather than a complexity constant at these scales. The two graph Laplacian engines are interchangeable; we use LAMG+ by default for its robustness on large graph instances.

5.5. Corpus Scaling

To confirm the near-linear exponent is a property of the method and not of the one size-scaled family, we time the equilibrium solve across a topology-diverse corpus of 96 real irregular graphs from the SuiteSparse/SNAP, LAW, DIMACS10, and Newman collections—autonomous-system, peer-to-peer, web-crawl, citation, collaboration, email, signed-social, and machine-learning k-nearest-neighbour graphs—spanning 1.1×10^4 to 6.0×10^6 arcs (2.74 decades). Timing was serial and uncontended (single run; build-net seed fixed). Every instance reaches a relative residual below 10^{-2} within a fixed per-graph budget (median 7.3×10^{-5}, with the inner-engine setup count in $[47, 330]$; that count correlates only weakly with size (log–log Pearson $r = 0.20$), so it is driven by conditioning, not m). The full-corpus wall-clock fits $t \propto m^{1.10}$ (OLS on log–log; 95% bootstrap CI $[1.02, 1.20]$, $R^2 = 0.85$)—near-linear, mildly super-linear. Restricting to the $m \geq 10^5$ subset (56 graphs) gives $t \propto m^{0.97}$. This corpus power law is descriptive, not a complexity proof: at a fixed size the solve time varies several-fold with conditioning (a hub-dominated web crawl can cost several times a same-size collaboration graph), which broadens the scatter, and the six instances with residual above 10^{-3} are the most hub-dominated crawls, whose one-sided law induces a stiff near-null Jacobian mode (Section 3).

Bilevel toll design (Algorithm 2) therefore inherits a near-linear per-step cost on large irregular networks, in contrast with the superlinear per-step factorization of the sensitivity-analysis incumbents. On near-planar road networks this advantage does not appear at practical sizes—nested dissection is near-optimal there—which is why we frame the advantage on the communication-network regime.

5.6. Comparison with Differentiable-Bilevel (First-Order) Solvers

The learning-based approach to Stackelberg congestion design (Li et al. 2022, Goyal and Lamperski 2023) treats the lower-level equilibrium as a differentiable layer: it is solved by a first-order scheme (unrolled imitative-logit dynamics; entropic mirror descent) and differentiated by unrolling or implicit differentiation. Their released code is toy-scale and does not run on the 10^4–10^5-node irregular graphs here, so we reproduce the defining component—a first-order inner equilibrium

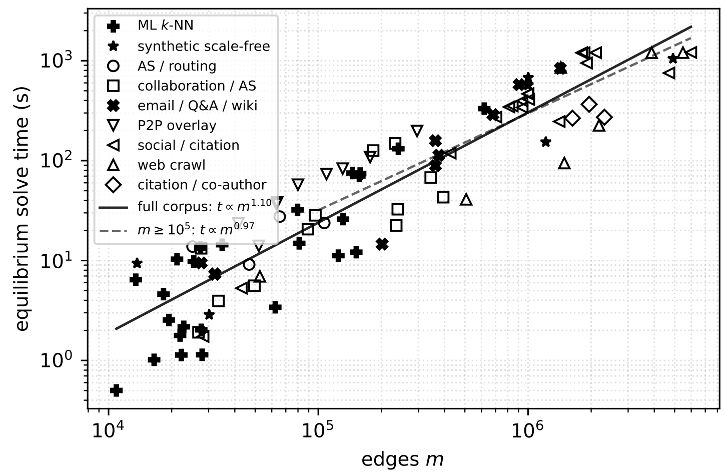

Figure 1 **Equilibrium solve time vs. edge count across a topology-diverse corpus of 96 real irregular graphs (SuiteSparse/SNAP, LAW, DIMACS10, Newman; autonomous-system, peer-to-peer, web-crawl, citation, collaboration, email, signed-social, and machine-learning k-NN families) spanning 1.1×10^4 to 6.0×10^6 arcs (2.74 decades). Full-corpus wall-clock fits $t \propto m^{1.10}$ (95% bootstrap CI $[1.02, 1.20]$, $R^2 = 0.85$); the dashed line is the $m \geq 10^5$ subset ($m^{0.97}$). Both are descriptive least-squares fits, not a complexity bound: an $\Omega(m)$ task cannot be asymptotically sub-linear, so neither exponent should be read as literal sub-linear complexity. Inner-engine setup count is $[47, 330]$, weakly size-correlated ($r = 0.20$). Residual median 7.3×10^{-5}, max 2.7×10^{-3}—a descriptive near-linear trend across real graphs of every family, not a complexity proof (Section 5).**

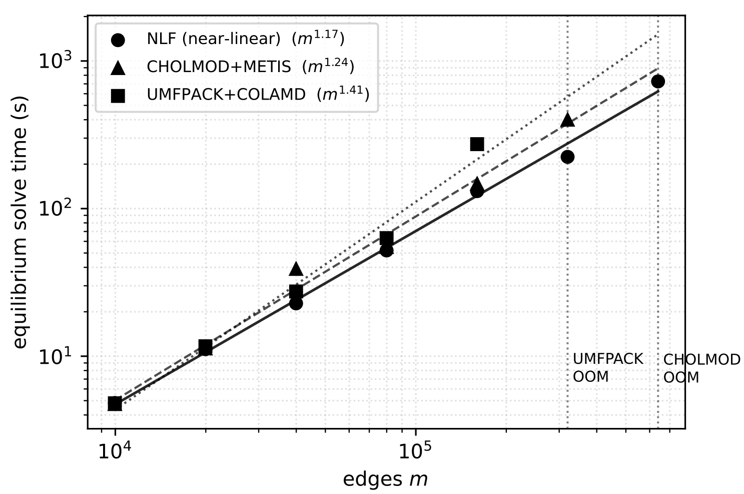

Figure 2 **Equilibrium solve time vs. edge count on size-scaled irregular congested networks: near-linear NLF ($m^{1.17}$) vs. the two direct factorizations of the SPD Laplacian—UMFPACK/COLAMD ($m^{1.41}$) and the fair, stronger CHOLMOD+METIS ($m^{1.24}$)—each of which exhausts memory (dotted walls) where NLF continues (Table 1).**

Table 1 Equilibrium solve time (s) on size-scaled irregular congested networks (random degree-5 graphs), one clean same-run head-to-head on an Apple M5 Pro; every completed solve reaches a 10^{-9} residual. The near-linear NLF engine (approximate Cholesky; the default LAMG+ tracks it within $\approx 1.5\times$) against the two direct factorizations of the pinned Laplacian J: the unsymmetric UMFPACK/COLAMD LU, and the fair, stronger CHOLMOD supernodal Cholesky under a METIS nested-dissection ordering (Proposition 1 makes J SPD). NLF fits $m^{1.17}$ and completes throughout; UMFPACK $m^{1.41}$, out of memory (OOM) at $m \approx 3.2 \times 10^5$; CHOLMOD+METIS $m^{1.24}$ overall ($m^{1.43}$ for $m \geq 8 \times 10^4$), reaching twice as far before OOM at $m \approx 6.4 \times 10^5$. The fair direct baseline is superlinear and hits a memory wall where NLF does not.

n	m	NLF (near-linear)	UMFPACK +COLAMD	CHOLMOD +METIS
1,000	9,964	4.82	4.80	4.74
2,000	19,946	11.16	11.68	11.29
4,000	39,952	22.80	27.45	39.03
8,000	79,936	52.04	63.12	54.96
16,000	159,954	131.32	273.26	147.07
32,000	319,952	224.10	*OOM*	400.21
64,000	639,950	727.01	*OOM*	*OOM*

solve—in the same environment and on the same instances. To isolate the inner solver, both methods minimize the identical convex Beckmann energy from the same cold start; only the inner iteration differs. For the first-order inner we use FISTA (accelerated proximal gradient with backtracking line search)—a strong, well-tuned member of the first-order family, and faster than plain mirror descent—so the comparison is not a strawman. The results are in Table 2. Our Newton inner reaches relative residual 10^{-9} in under two minutes on every instance; FISTA, given a 300-second budget on every instance, never reaches the 10^{-6} target. Its final residual is governed by conditioning, not size: it reaches $\sim 10^{-4}$–10^{-5} on the well-conditioned synthetic family (Table 2) but only 2.4×10^{-3} to 5.2×10^{-2} on the stiffer real-topology instances. It approaches the same solution in flow but never to solve accuracy. The obstruction is fundamental, not a tuning artifact: heavy congestion makes the Beckmann Hessian severely ill-conditioned, and an accelerated first-order method still needs $\Theta(\sqrt{\kappa})$ iterations, so on the largest real instance (`as-caida`, $m = 1.07 \times 10^5$) 300 s of FISTA reaches only 5.2×10^{-2}—seven orders short of the Newton inner's 10^{-9}. A first-order inner therefore has no fast, size-independent convergence on stiff equilibria, whereas the Newton inner does—which is why these methods are demonstrated only at small scale, and why replacing their inner solve is novel.

5.7. Reproducibility

All code, unit tests, and reproduction drivers are available at `https://github.com/orenlivne/dnlf`; the repository regenerates every table and figure in this paper—the widening advantage

Table 2 **Inner-solver head-to-head on the identical convex equilibrium, same cold start. Ours (Newton + AC, fine continuation) reaching relative residual 10^{-9}; FISTA (accelerated first-order—the differentiable-bilevel inner) targeting 10^{-6} but capped at 300 s, which it hits on every instance, reaching only the residual shown.**

instance	n	m	ours (s)	FISTA (s)	FISTA relres
`synthetic`	1,000	9,964	3.9	300	1.2×10^{-4}
`synthetic`	2,000	19,946	9.8	300	2.4×10^{-5}
`synthetic`	4,000	39,952	20.0	300	1.0×10^{-5}
`p2p-Gnutella08`	6,299	41,552	29.6	300	2.4×10^{-3}
`Oregon-1`	11,174	46,818	50.5	300	1.6×10^{-2}
`as-caida`	26,475	106,762	82.9	300	5.2×10^{-2}

Table 3 **Reproducibility: solver parameters and their repository flags. Accurate mode (ε_{int} disabled) is used for the correctness and design results (Sections 4 to 5); the loose-homotopy, adaptive-polish mode for the corpus (Fig. 1) and the design-at-scale run.**

symbol	meaning	value	repo flag
ε_{tol}	final/accurate tolerance	10^{-9}	`tol`
ε_{int}	loose intermediate tolerance	3×10^{-2}, ≤ 6 steps	`itol`, `inmax`
δ_k	δ-schedule (ratio, count)	$\frac{1}{2}\bar{t}^0\, 0.85^{k-1}$, $k\leq 111$	`dfracs`
ε	relative conductance floor	10^{-12}	(fixed)
ε_{amg}	inner AMG/PCG tolerance	5×10^{-2}	`eta`
—	loose homotopy, adaptive polish	on	`tight_last=false`
—	backend (Tables 1 and 2)	LAMG+/AC/LU/CHOLMOD	`inner`
γ	logit dispersion (target)	0.1; 0.15; 2, 5 (roads)	`gamma`
γ_0	γ-continuation from	2.0 ($\times\frac{1}{2}$; tol 10^{-8})	`gamma0`
η_0	design step; ≤ 20, $\times\frac{1}{2}$	$0.5\,\bar{\tau}_{\text{Pig}}/\|g\|_\infty$	`design_step`
seed	graph/demand RNG	1	`MersenneTwister`

(Table 1, Fig. 2), the corpus scaling (Fig. 1), the first-order head-to-head (Table 2), and the multicommodity tables (Tables 4 to 6)—and unit-tests every algorithm component, including Propositions 1 to 4 and the OD-matrix loader. All timings reported in this paper were measured on an Apple M5 Pro (18 cores, 48 GB RAM) under macOS 26.5. The test networks are the TNTP Sioux Falls and Anaheim instances with their published OD matrices (shipped with the code) and public Internet AS / P2P-overlay graphs (e.g. `as-caida`, `Oregon-1`, `p2p-Gnutella`) from the SuiteSparse/SNAP collection. Table 3 pins every solver parameter to a symbol and a repository flag. The synthetic instances (Tables 1, 2 and 5) use random degree-5 graphs and a demand of $\lceil n/8 \rceil$ random source/sink pairs scaled to a fixed total, all under the fixed seed of Table 3; each corpus instance places the same synthetic demand and BPR parameters on the real topology (so Fig. 1 times a fixed, documented congestion problem, not an ad-hoc one). Because approximate Cholesky is randomized, reported iteration/setup counts are from that fixed seed; the repository reproduces them exactly and reports run-to-run spread.

6. Multicommodity logit stochastic user equilibrium and design

The single-commodity solver of Section 3 routes a general balanced demand as one fungible flow. Realistic congestion pricing is inherently multicommodity: each origin–destination (OD) pair routes its own flow over the shared congested network, and tolls are designed against that coupled equilibrium. We extend the framework to the multicommodity *logit stochastic user equilibrium* (SUE) (Fisk 1980, Daganzo and Sheffi 1977), in which route choice has a dispersion $\gamma > 0$—a calibrated behavioural parameter, not a numerical smoothing. The same near-linear machinery carries over: the multicommodity Newton system is a coupled block-Laplacian that a block-diagonal preconditioner solves in a handful of iterations, and the design adjoint again returns $\nabla_\tau \mathrm{TSTT}$ for all tolls in a single solve.

6.1. Formulation

With K commodities, per-commodity arc flows $f^k \geq 0$, total flow $f_a = \sum_k f_a^k$, and the shared cost $t_a(f_a)$, the entropic-regularized Beckmann program is

$$\min_{f^k \geq 0} \sum_a T_a(f_a) + \gamma \sum_{a,k} f_a^k \big(\log f_a^k - 1\big) \quad \text{s.t.} \quad B f^k = d^k, \quad k = 1, \dots, K, \tag{5}$$

with $T_a' = t_a$. Its stationarity conditions give the logit assignment $f_a^k = \exp\!\big((g_a^k - t_a(f_a))/\gamma\big)$ with $g_a^k = (B^\top \phi^k)_a$: a per-arc softmax over commodities whose total f_a solves a monotone scalar fixed point $\log f_a = \mathrm{logsumexp}_k(g_a^k/\gamma) - t_a(f_a)/\gamma$. As $\gamma \to 0$ the model tends to deterministic user equilibrium; we solve at a fixed, calibrated γ.

Proposition 4 **(multicommodity Newton system: a coupled block Laplacian)**. *At a logit equilibrium the reduced Newton Jacobian in the stacked potentials* Φ *is*

$$J = \underbrace{\tfrac{1}{\gamma}\, \mathrm{blkdiag}_k\big(B\, \mathrm{diag}(f^k)\, B^\top\big)}_{K\ \textit{per-commodity Laplacians}} - \tfrac{1}{\gamma^2} \sum_a c_a\, (\mathbf{f}_a \mathbf{f}_a^\top) \otimes b_a b_a^\top, \tag{6}$$

where $\Phi = [\phi^1; \dots; \phi^K]$, $b_a = B e_a$, $\mathbf{f}_a = (f_a^1, \dots, f_a^K)^\top$ *(the per-commodity vector; the scalar total is* $f_a = \sum_k f_a^k$*), and* $c_a = t_a'/(1 + t_a' f_a/\gamma)$*. Each diagonal block is a symmetric weighted graph Laplacian; the coupling is symmetric positive semidefinite and rank one per arc. For* $K = 1$*, Eq.* (6) *reduces, as* $\gamma \to 0$*, to the single-commodity active Laplacian* $B\, \mathrm{diag}(1/t_a')\, B^\top$ *of Proposition 1.*

Proof. The equilibrium is $R^k(\Phi) = B f^k(\Phi) - d^k = 0$ (matching the constraint $B f^k = d^k$). Differentiating the logit map, $\mathrm{d}f_a^k = (f_a^k/\gamma)(\mathrm{d}g_a^k - t_a'\, \mathrm{d}f_a)$ and $\mathrm{d}f_a = \sum_l \mathrm{d}f_a^l$; solving the latter gives $\mathrm{d}f_a = (\gamma + t_a' f_a)^{-1} \sum_l f_a^l\, \mathrm{d}g_a^l$. Substituting and using $\mathrm{d}g^k = B^\top \mathrm{d}\phi^k$, $\mathrm{d}R^k = B\, \mathrm{d}f^k$ yields Eq. (6). Equivalently $J = \tilde{B} H^{-1} \tilde{B}^\top$ with $\tilde{B} = I_K \otimes B$ and per-arc flow-Hessian $H_a = t_a' \mathbf{1}\mathbf{1}^\top + \gamma\, \mathrm{diag}(1/f_a^k)$; Sherman–Morrison gives $H_a^{-1} = \frac{1}{\gamma} \mathrm{diag}(f_a^k) - \frac{c_a}{\gamma^2} \mathbf{f}_a \mathbf{f}_a^\top$, and for $K = 1$, $f_a/(\gamma + t_a' f_a) \to 1/t_a'$ as $\gamma \to 0$. Q.E.D.

6.2. Near-Linear Solve

We solve Eq. (5) by γ-continuation damped Newton on Φ, preconditioning Eq. (6) by its block diagonal—one near-linear (AC) solve per commodity. The block diagonal misses only the rank-one coupling of Eq. (6), through which the commodities interact solely via the shared total flow $f_a = \sum_k f_a^k$ that sets each arc's cost. In the commodity mean/deviation basis this coupling sits on the single mean mode: when the per-commodity Laplacians coincide the $K-1$ deviation modes decouple from it exactly, leaving only that one mode for preconditioned CG to resolve; when the Laplacians differ the deviations mix in as well, at a rate set only by their spread. Empirically (Table 4) the number of conjugate-gradient iterations ≤ 13, grows only logarithmically in K, and is independent of graph size n at the operating dispersion $\gamma \approx 0.1$; each Newton step therefore costs $O(K)$ near-linear solves, and the equilibrium solve is near-linear in the total size Kn. Each per-commodity Laplacian carries a small relative conductance floor (10^{-12} of its maximum weight) so it stays connected and the AC factorization remains applicable even to a commodity that concentrates on few arcs. On the rare block where AC throws—a commodity so concentrated that nearly all arcs sit at the floor, leaving a near-disconnected, high-contrast Laplacian that Laplacians.jl's randomized elimination can zero-pivot on, which happens only in the cold far-from-equilibrium regime—we fall back to a direct solve; this changes only the preconditioner, not the PCG-converged solve. The inner count stays single-digit at and near equilibrium at the operating dispersion $\gamma \approx 0.1$—the regime the warm-started synthetic design loop lives in (Table 5)—and low double-digit (≤ 38) on the small, heavily loaded road instances at their higher calibrated γ (Table 6). Only the *cold* solve at small γ, far from equilibrium where per-commodity flows concentrate into high-contrast blocks, is expensive, and that regime is approached only as $\gamma \to 0$—a different, harder model (deterministic multicommodity UE, with non-unique flows) already served by Frank–Wolfe. Since fixed γ is a calibrated route-choice dispersion (Fisk 1980, Daganzo and Sheffi 1977), a behavioural model in its own right, we solve the logit SUE at that γ and need not pursue $\gamma \to 0$. (We also tested a coupling-aware preconditioner—per-arc rank-one coupling inverted by Sherman–Morrison, aggregate-congestion Laplacian on the mean commodity mode: it converges γ-boundedly where the block-diagonal rate degrades, but does not beat the block diagonal in the operating regime, so the block diagonal is retained.)

6.3. Design Optimization

Tolls enter the logit map as a per-arc cost shift $t_a(f_a) + \tau_a$, with flow sensitivity $\partial f_a^k / \partial \tau_a = -f_a^k c_a$. The same adjoint as Section 4 returns $\nabla_\tau \mathrm{TSTT}$ for all tolls from a single block-preconditioned solve $J\lambda = \partial_\Phi \mathrm{TSTT}$. The adjoint matches central finite differences to under 10^{-3} (verified by the released unit tests), and on a multicommodity instance projected-gradient tolling reduces TSTT

Table 4 **Multicommodity logit-SUE inner conjugate-gradient iterations (block-preconditioned) on irregular networks, $\gamma = 0.1$. Left: vs. commodities K ($n = 1500$), single-OD and spread demands. Right: vs. graph size n ($K = 8$, spread). Single-digit to low double digit, $\log K$ growth, n-independent.**

K	single-OD	spread	n (at K=8)	iters
2	3	2	1,000	6
4	3	3	2,000	6
8	7	6	4,000	7
12	13	10	8,000	3
16	12	9	16,000	4

with the tolls spread across the congested arcs, each equilibrium re-solve costing single-digit inner iterations—the bilevel iteration inheriting the near-linear per-step cost. Across a range of synthetic instances (varying K, γ, network size, and demand; Table 5) the reduction is 1–2% and the inner-iteration count stays single digit. On the two canonical TNTP road networks with their *published* origin–destination matrices and BPR parameters—Sioux Falls ($n = 24$, $m = 76$, $K = 24$ destination commodities) and Anaheim ($n = 416$, $m = 914$, $K = 38$)—destination-bundled design reduces the real system travel time by 7.7–13.1% (Table 6), at the higher dispersions $\gamma \in \{2, 5\}$ these small over-saturated road instances require for a well-conditioned logit SUE (the $\gamma \approx 0.1$ operating point of the size-independence study yields a stiffer block system on so few, so heavily loaded arcs); the reduction accordingly varies with γ and should be read as a γ-dependent span, not a single design gain. Both networks are over-saturated at their full published demand (peak volume/capacity above 2.5); the inner iteration count stays bounded and does not grow with size—though here it is low double digit (12–38), not single digit, again because higher γ and heavy loading strain the block-diagonal preconditioner (consistent with the size-independence, but not the magnitude, of the synthetic family, Table 4). What sets the count is not size but the per-arc commodity coupling the block-diagonal preconditioner omits, which is significant only where arcs approach the BPR saturation knee: tiny, dense Sioux Falls at its over-saturated published demand forces most carrying arcs there, whereas Anaheim spreads flow so typical arcs stay clear of it. A budget-constrained (sparse-toll) design is a natural extension.

7. Conclusions, limitations, and open problems

We showed that the directed user equilibrium with separable costs is a nonlinear Laplacian flow whose Jacobian is a symmetric active-subgraph Laplacian, so a smoothing homotopy reduces the forward solve and the adjoint design gradient to a short sequence of near-linear Laplacian solves with a graph-size-independent number of multigrid setups; bilevel toll design inherits this near-linear per-step cost, in place of the superlinear (fill-dominated) factorization of sensitivity-analysis methods.

Table 5 **Multicommodity toll design across instances (varying commodities K, dispersion γ, network size n, and per-commodity demand “load”): TSTT reduction, inner conjugate-gradient iterations per equilibrium re-solve, and the fraction of arcs carrying a non-negligible toll ($\tau > 10^{-3}$ of the maximum). The inner count stays single digit throughout.**

n	K	γ	load	TSTT red.	inner it.	arcs tolled
800	4	0.10	3,000	1.1%	3	29%
800	8	0.10	3,000	1.1%	4	49%
800	16	0.10	3,000	1.2%	7	67%
800	8	0.15	3,000	1.3%	3	72%
800	8	0.10	6,000	1.0%	6	49%
800	8	0.10	3,200	1.4%	6	45%
1,200	8	0.10	3,600	1.4%	6	44%
1,600	8	0.10	3,600	1.2%	6	43%

Table 6 **Real-network validation: destination-bundled logit-SUE toll design on the two canonical TNTP road networks with their published OD matrices and BPR parameters (one commodity per destination zone). Both are over-saturated at full demand (peak volume/capacity > 2.5). Tolls spread across 53–74% of arcs. The inner conjugate-gradient count at the final equilibrium solve is bounded (12–38); it is set by proximity to the BPR saturation knee (Section 6), not by network size.**

network	n	m	K	γ	TSTT red.	inner it.
Sioux Falls	24	76	24	2	9.4%	38
Sioux Falls	24	76	24	5	13.1%	30
Anaheim	416	914	38	2	12.2%	16
Anaheim	416	914	38	5	7.7%	12

We are explicit about what this paper does not yet establish. (i) **Multicommodity scope.** Section 6 solves and designs the multicommodity logit SUE at fixed calibrated γ; we do not pursue the $\gamma \to 0$ deterministic limit, a distinct model (non-unique per-commodity flows) already served by Frank–Wolfe. The multicommodity design is demonstrated at road-network scale (Sioux Falls, Anaheim); the metropolitan-scale design demonstration (Section 4) is single-commodity, so a multicommodity design at that scale remains to be shown. Reaching many commodities is limited by the $\mathcal{O}(Km)$ cost of the K per-commodity block solves per step; since the commodities couple only through the low-rank, shared-arc congestion term (Proposition 4), a promising direction is to aggregate destinations of similar arc-footprint into $K' \ll K$ *super-commodities* used as a deflation space for the block-Newton system, reducing the effective block-solve count toward the numerical rank of the coupling where footprints cluster (to be assessed against specialized multicommodity interior-point solvers (Castro 2000)). (ii) **Accuracy on the most hub-dominated graphs.** On a handful of extreme hub graphs (dense web crawls) the one-sided law induces a stiff near-null Jacobian mode that the adaptive-polish solve resolves only to $\sim 10^{-3}$ within budget rather than machine precision (Section 3); this suffices for design (which needs $\sim 10^{-3}$) but a mode-deflating polish is future work. (iii) **One hierarchy across design steps.** Each exact solve rebuilds once per δ-level because crossing an active-set boundary raises a near-null-space mode; capturing that mode by deflation or recombination would make one hierarchy valid across all design steps and cut the setups to $\mathcal{O}(1)$ *total*—the main open algorithmic problem. (iv) **Planarity.** The advantage is on non-planar, poorly-separable topologies; on near-planar road networks nested dissection is already near-optimal (Section 1). (v) **Saturating laws and folds.** A capacity-saturating (min-delay) law reintroduces a turning-point fold; design through the fold, via the continuation/deflation of Livne (2026b), is future work.

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